



Certificate of Online Course Participation

I hereby certify that

Mrs. Brisa Oliveira e Silva Bøhle

has attended the **Ladybug tools online workshop** on October 17th through October 19th,
2022.

The workshop was taught by the cofounders of Dosta Tec, **Adrian Krężlik** and **Mateo Barbero**, and **Dr. Emanuele Naboni**, Associate Professor at the University of Parma, Italy. It was organised by **McNeel Europe**. The three-day online course introduced **Ladybug tools**, a free Grasshopper plugin to support **parametric environmental design and energy analysis**. Participants learned theoretical and practiced based fundamentals of energy modelling using Ladybug tools with a focus on the creation of energy intensity simulations, indoor thermal comfort, and daylight autonomy. Standards such as LEED and BREEAM were taken into consideration.

The online course covered 18 hours.

Barcelona, November 11th 2022

Verena Vogler,

Training and R&D department at McNeel Europe